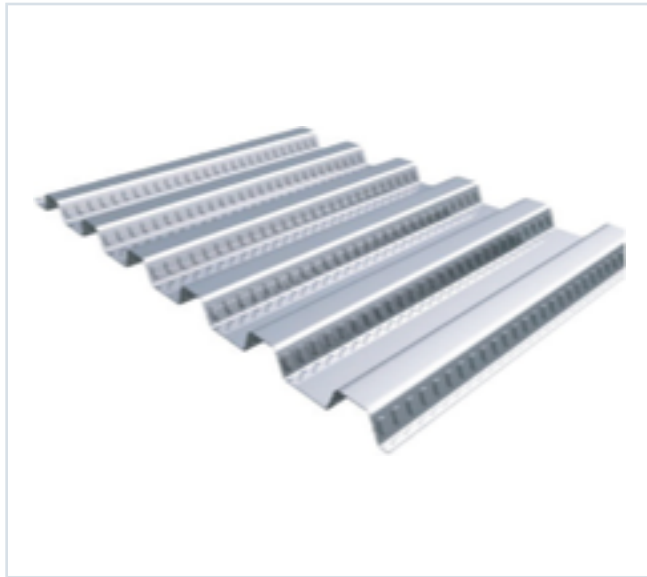




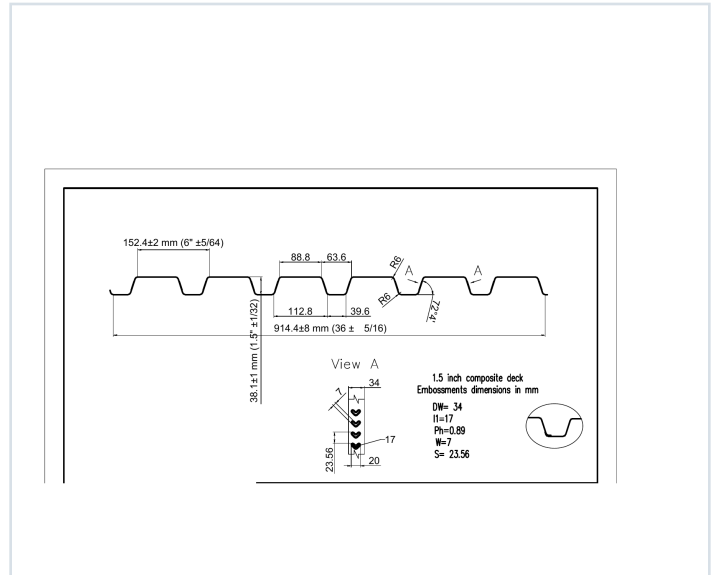
# DECKING PRODUCT SUBMITTAL

| PRODUCT             | GRADE    | GAUGE | COATING |
|---------------------|----------|-------|---------|
| 1.5" Composite Deck | Grade 50 | 18 ga | G60     |

## PRODUCT VIEW



## PROFILE DRAWING



## SECTION PROPERTIES · 18 GAUGE

| Thickness<br>(in.) | Weight<br>(psf) | Fy<br>(ksi) | S+<br>(in³/ft) | S-<br>(in³/ft) | M+<br>(in-lb/ft) | M-<br>(in-lb/ft) | I+<br>(in■/ft) | I-<br>(in■/ft) |
|--------------------|-----------------|-------------|----------------|----------------|------------------|------------------|----------------|----------------|
| 0.0474             | 2.6             | 50          | 0.294          | 0.310          | 8812             | 9291             | 0.257          | 0.290          |

## SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 45, 46, 47, 48
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

# PUBLISHED PERFORMANCE DATA

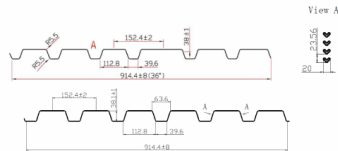
15CD-FY50-18-G60 · 18 GAUGE

## OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

### STEEL CATALOG PAGE 45

GRADE 50 STEEL



#### Section Properties

| Gage | Design Thickness (inches) | Weight (lb/ft) | F <sub>y</sub> (ksi) | F <sub>t</sub> (ksi) | S <sub>x</sub> (in <sup>3</sup> ) | S <sub>y</sub> (in <sup>3</sup> ) | ASD (S <sub>x</sub> + 147) | I <sub>x</sub> (in <sup>4</sup> ) | I <sub>y</sub> (in <sup>4</sup> ) |
|------|---------------------------|----------------|----------------------|----------------------|-----------------------------------|-----------------------------------|----------------------------|-----------------------------------|-----------------------------------|
| 22   | 0.0296                    | 16             | 50                   | 0.770                | 0.222                             | 0.001                             | 558                        | 0.144                             | 0.107                             |
| 20   | 0.0368                    | 20             | 50                   | 0.294                | 0.330                             | 0.001                             | 558                        | 0.182                             | 0.290                             |
| 18   | 0.0474                    | 26             | 50                   | 0.294                | 0.330                             | 0.001                             | 558                        | 0.257                             | 0.290                             |
| 16   | 0.0598                    | 33             | 50                   | 0.294                | 0.330                             | 0.001                             | 558                        | 0.341                             | 0.363                             |

Note: All section properties and ASD/bearing strengths are calculated in accordance with AISI/S10-2017, AISI S100-2012 and AISI S100-2016.

#### Shear and Web Crippling

| Gage | V <sub>u</sub> /Q (lb/ft) | Web Crippling (R <sub>u</sub> /Q), lbs/ft One Flange Loading End Bearing | Web Crippling (R <sub>u</sub> /Q), lbs/ft One Flange Loading Interior Bearing |
|------|---------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 22   | 2424                      | 1015                                                                     | 1015                                                                          |
| 20   | 3803                      | 1535                                                                     | 1535                                                                          |
| 18   | 5052                      | 2077                                                                     | 2077                                                                          |
| 16   | 6279                      | 2597                                                                     | 2597                                                                          |

Note: All section properties and ASD/bearing strengths are calculated in accordance with AISI/S10-2017, AISI S100-2012 and AISI S100-2016.

#### Allowable Uniform Downward Loads, ASD (PSF)

| Span   | Gage | 5'-0" | 5'-6" | 6'-0" | 6'-6" | 7'-0" | 7'-6" | 8'-0" | 8'-6" | 9'-0" | 9'-6" | 10'-0" |
|--------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
| Single | 22   | 158   | 12    | 94    | 80    | 69    | 60    | 53    | 47    | 42    | 38    | 34     |
|        | 20   | 272   | 142   | 100   | 82    | 71    | 62    | 55    | 49    | 44    | 40    | 36     |
|        | 18   | 235   | 124   | 85    | 71    | 62    | 55    | 49    | 44    | 40    | 36    | 32     |
|        | 16   | 202   | 107   | 75    | 62    | 55    | 49    | 44    | 40    | 36    | 32    | 28     |
| Double | 22   | 143   | 118   | 89    | 75    | 65    | 56    | 49    | 44    | 40    | 36    | 32     |
|        | 20   | 198   | 147   | 125   | 105   | 91    | 79    | 69    | 61    | 55    | 49    | 44     |
|        | 18   | 246   | 185   | 132   | 110   | 95    | 83    | 73    | 65    | 59    | 53    | 47     |
|        | 16   | 291   | 227   | 164   | 139   | 120   | 105   | 93    | 83    | 75    | 67    | 60     |
| Triple | 22   | 179   | 148   | 104   | 86    | 74    | 64    | 56    | 49    | 44    | 40    | 36     |
|        | 20   | 262   | 185   | 134   | 111   | 96    | 83    | 73    | 65    | 59    | 53    | 47     |
|        | 18   | 230   | 165   | 116   | 95    | 83    | 73    | 65    | 59    | 53    | 47    | 42     |
|        | 16   | 309   | 231   | 165   | 134   | 116   | 100   | 87    | 77    | 69    | 62    | 55     |

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

### STEEL CATALOG PAGE 46

| Span   | Gage | 5'-0" | 5'-6" | 6'-0" | 6'-6" | 7'-0" | 7'-6" | 8'-0" | 8'-6" | 9'-0" | 9'-6" | 10'-0" |
|--------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
| Single | 22   | 16    | 12    | 44    | 35    | 29    | 22    | 19    | 15    | 13    | 11    | 9      |
|        | 20   | 16    | 12    | 36    | 44    | 35    | 28    | 23    | 19    | 16    | 14    | 12     |
|        | 18   | 13    | 12    | 78    | 61    | 49    | 40    | 33    | 27    | 23    | 20    | 17     |
|        | 16   | 179   | 135   | 104   | 82    | 65    | 53    | 44    | 36    | 31    | 26    | 22     |
| Double | 22   | 133   | 107   | 106   | 83    | 67    | 54    | 46    | 37    | 31    | 27    | 23     |
|        | 20   | 230   | 173   | 133   | 105   | 84    | 68    | 56    | 47    | 40    | 34    | 29     |
|        | 18   | 325   | 244   | 188   | 148   | 118   | 96    | 79    | 66    | 56    | 47    | 41     |
|        | 16   | 431   | 324   | 250   | 196   | 157   | 128   | 105   | 88    | 74    | 63    | 54     |
| Triple | 22   | 143   | 107   | 83    | 65    | 52    | 42    | 35    | 29    | 25    | 21    | 18     |
|        | 20   | 180   | 135   | 104   | 82    | 66    | 53    | 44    | 37    | 31    | 26    | 22     |
|        | 18   | 254   | 191   | 147   | 116   | 95    | 75    | 62    | 52    | 44    | 37    | 32     |
|        | 16   | 338   | 254   | 195   | 154   | 123   | 100   | 82    | 69    | 58    | 49    | 42     |

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

#### Construction Span Table - 20 psf Construction Load

| Normal Weight Concrete (145 pcf)           |            |                             |         |         |                                            | Lightweight Concrete (115 pcf) |                             |         |        |  |  |
|--------------------------------------------|------------|-----------------------------|---------|---------|--------------------------------------------|--------------------------------|-----------------------------|---------|--------|--|--|
| Total Slab Depth<br>in<br>(#500<br>31 PSF) | Deck Type  | Maximum Unstayed Clear Span |         |         | Total Slab Depth<br>in<br>(#500<br>31 PSF) | Deck Type                      | Maximum Unstayed Clear Span |         |        |  |  |
|                                            |            | 1 span                      | 2 span  | 3 span  |                                            |                                | 1 span                      | 2 span  | 3 span |  |  |
| 400<br>(#500)<br>37 PSF                    | 15x6x22 ga | 6' 0"                       | 7' 0"   | 7' 9"   | 400<br>(#500)<br>37 PSF                    | 15x6x22 ga                     | 7' 3"                       | 8' 7"   | 9' 8"  |  |  |
|                                            | 15x6x18 ga | 7' 0"                       | 8' 12"  | 9' 11"  |                                            | 15x6x18 ga                     | 8' 0"                       | 9' 12"  | 10' 4" |  |  |
|                                            | 15x6x16 ga | 8' 0"                       | 9' 12"  | 10' 11" |                                            | 15x6x16 ga                     | 9' 0"                       | 10' 12" | 11' 4" |  |  |
|                                            | 15x6x14 ga | 9' 0"                       | 10' 12" | 11' 4"  |                                            | 15x6x14 ga                     | 10' 0"                      | 11' 4"  | 12' 4" |  |  |
|                                            | 15x6x12 ga | 7' 3"                       | 8' 6"   | 8' 7"   |                                            | 15x6x12 ga                     | 8' 2"                       | 9' 7"   | 9' 8"  |  |  |
| 450<br>(#500)<br>43 PSF                    | 15x6x22 ga | 6' 0"                       | 7' 0"   | 7' 9"   | 450<br>(#500)<br>43 PSF                    | 15x6x22 ga                     | 7' 3"                       | 8' 7"   | 9' 8"  |  |  |
|                                            | 15x6x18 ga | 7' 0"                       | 8' 12"  | 9' 11"  |                                            | 15x6x18 ga                     | 8' 0"                       | 9' 12"  | 10' 4" |  |  |
|                                            | 15x6x16 ga | 8' 0"                       | 9' 12"  | 10' 11" |                                            | 15x6x16 ga                     | 9' 0"                       | 10' 12" | 11' 4" |  |  |
|                                            | 15x6x14 ga | 9' 0"                       | 10' 12" | 11' 4"  |                                            | 15x6x14 ga                     | 10' 0"                      | 11' 4"  | 12' 4" |  |  |
|                                            | 15x6x12 ga | 5' 0"                       | 6' 0"   | 7' 0"   |                                            | 15x6x12 ga                     | 6' 0"                       | 7' 0"   | 7' 9"  |  |  |
| 500<br>(#500)<br>49 PSF                    | 15x6x22 ga | 6' 0"                       | 7' 0"   | 7' 9"   | 500<br>(#500)<br>49 PSF                    | 15x6x22 ga                     | 7' 3"                       | 8' 7"   | 9' 8"  |  |  |
|                                            | 15x6x18 ga | 7' 0"                       | 8' 12"  | 9' 11"  |                                            | 15x6x18 ga                     | 8' 0"                       | 9' 12"  | 10' 4" |  |  |
|                                            | 15x6x16 ga | 8' 0"                       | 9' 12"  | 10' 11" |                                            | 15x6x16 ga                     | 9' 0"                       | 10' 12" | 11' 4" |  |  |
|                                            | 15x6x14 ga | 9' 0"                       | 10' 12" | 11' 4"  |                                            | 15x6x14 ga                     | 10' 0"                      | 11' 4"  | 12' 4" |  |  |
|                                            | 15x6x12 ga | 6' 0"                       | 7' 0"   | 7' 9"   |                                            | 15x6x12 ga                     | 7' 0"                       | 8' 0"   | 8' 7"  |  |  |
| 550<br>(#500)<br>55 PSF                    | 15x6x22 ga | 6' 0"                       | 7' 0"   | 7' 9"   | 550<br>(#500)<br>55 PSF                    | 15x6x22 ga                     | 7' 3"                       | 8' 7"   | 9' 8"  |  |  |
|                                            | 15x6x18 ga | 7' 0"                       | 8' 12"  | 9' 11"  |                                            | 15x6x18 ga                     | 8' 0"                       | 9' 12"  | 10' 4" |  |  |
|                                            | 15x6x16 ga | 8' 0"                       | 9' 12"  | 10' 11" |                                            | 15x6x16 ga                     | 9' 0"                       | 10' 12" | 11' 4" |  |  |
|                                            | 15x6x14 ga | 9' 0"                       | 10' 12" | 11' 4"  |                                            | 15x6x14 ga                     | 10' 0"                      | 11' 4"  | 12' 4" |  |  |
|                                            | 15x6x12 ga | 5' 0"                       | 6' 0"   | 6' 6"   |                                            | 15x6x12 ga                     | 6' 0"                       | 7' 0"   | 7' 9"  |  |  |
| 600<br>(#500)<br>61 PSF                    | 15x6x22 ga | 6' 0"                       | 7' 0"   | 7' 9"   | 600<br>(#500)<br>61 PSF                    | 15x6x22 ga                     | 7' 3"                       | 8' 7"   | 9' 8"  |  |  |
|                                            | 15x6x18 ga | 7' 0"                       | 8' 12"  | 9' 11"  |                                            | 15x6x18 ga                     | 8' 0"                       | 9' 12"  | 10' 4" |  |  |
|                                            | 15x6x16 ga | 8' 0"                       | 9' 12"  | 10' 11" |                                            | 15x6x16 ga                     | 9' 0"                       | 10' 12" | 11' 4" |  |  |
|                                            | 15x6x14 ga | 9' 0"                       | 10' 12" | 11' 4"  |                                            | 15x6x14 ga                     | 10' 0"                      | 11' 4"  | 12' 4" |  |  |
|                                            | 15x6x12 ga | 6' 0"                       | 7' 0"   | 7' 9"   |                                            | 15x6x12 ga                     | 6' 0"                       | 7' 0"   | 7' 9"  |  |  |

Note: Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

# PUBLISHED PERFORMANCE DATA

15CD-FY50-18-G60 · 18 GAUGE

## OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

### STEEL CATALOG PAGE 47

| LSC Deck - Wide Flange Up |            |                                 |                                 | SINV Deck - Wide Flange Down |            |                                 |                                 |
|---------------------------|------------|---------------------------------|---------------------------------|------------------------------|------------|---------------------------------|---------------------------------|
| 50 ksi                    |            |                                 |                                 | 50 ksi                       |            |                                 |                                 |
| Live Load                 | 20         | psf                             | with concrete                   | Live Load                    | 20         | psf                             | with concrete                   |
| DL                        | 160        | psf                             |                                 | DL                           | 160        | psf                             |                                 |
| Deck Weight               | 2          | psf                             |                                 | Deck Weight                  | 2          | psf                             |                                 |
| Live Load                 | 50         | psf                             | without concrete                | Live Load                    | 50         | psf                             | without concrete                |
| Total Slab Thickness      | Deck Gauge | Maximum Cantilever (inches) NWC | Maximum Cantilever (inches) LWC | Total Slab Thickness         | Deck Gauge | Maximum Cantilever (inches) NWC | Maximum Cantilever (inches) LWC |
| 3.5                       | 22         | 36                              | 36                              | 3.5                          | 22         | 25                              | 25                              |
| 3.5                       | 20         | 31                              | 31                              | 3.5                          | 20         | 30                              | 30                              |
| 3.5                       | 18         | 40                              | 40                              | 3.5                          | 18         | 38                              | 38                              |
| 3.5                       | 16         | 47                              | 47                              | 3.5                          | 16         | 46                              | 46                              |
| 4                         | 22         | 36                              | 36                              | 4                            | 22         | 25                              | 25                              |
| 4                         | 20         | 31                              | 31                              | 4                            | 20         | 30                              | 30                              |
| 4                         | 18         | 40                              | 40                              | 4                            | 18         | 38                              | 38                              |
| 4                         | 16         | 47                              | 47                              | 4                            | 16         | 46                              | 46                              |
| 4.5                       | 22         | 36                              | 36                              | 4.5                          | 22         | 25                              | 25                              |
| 4.5                       | 20         | 31                              | 31                              | 4.5                          | 20         | 30                              | 30                              |
| 4.5                       | 18         | 40                              | 40                              | 4.5                          | 18         | 38                              | 38                              |
| 4.5                       | 16         | 47                              | 47                              | 4.5                          | 16         | 46                              | 46                              |
| 5                         | 22         | 36                              | 36                              | 5                            | 22         | 25                              | 25                              |
| 5                         | 20         | 31                              | 31                              | 5                            | 20         | 30                              | 30                              |
| 5                         | 18         | 40                              | 40                              | 5                            | 18         | 38                              | 38                              |
| 5                         | 16         | 47                              | 47                              | 5                            | 16         | 46                              | 46                              |
| 5.5                       | 22         | 36                              | 36                              | 5.5                          | 22         | 25                              | 25                              |
| 5.5                       | 20         | 31                              | 31                              | 5.5                          | 20         | 30                              | 30                              |
| 5.5                       | 18         | 40                              | 40                              | 5.5                          | 18         | 38                              | 38                              |
| 5.5                       | 16         | 47                              | 47                              | 5.5                          | 16         | 46                              | 46                              |
| 6                         | 22         | 36                              | 36                              | 6                            | 22         | 25                              | 25                              |
| 6                         | 20         | 31                              | 31                              | 6                            | 20         | 30                              | 30                              |
| 6                         | 18         | 40                              | 40                              | 6                            | 18         | 38                              | 38                              |
| 6                         | 16         | 47                              | 47                              | 6                            | 16         | 46                              | 46                              |

Assumes that backspan is greater than the cantilever spans.  
Applicable to single, double, triple spans or greater.  
Controlled by concentrated load = 30 psf  
Controlled by concrete weight = 30 psf  
Controlled by Deflection

### STEEL CATALOG PAGE 48

| 22 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi) |              |       |       |        |        |        |        |        |        |
|--------------------------------------------------------|--------------|-------|-------|--------|--------|--------|--------|--------|--------|
| Slab Thickness (inches)                                | Weight (psf) | 8'-0" | 9'-0" | 10'-0" | 11'-0" | 12'-0" | 13'-0" | 14'-0" | 15'-0" |
| 3.5                                                    | 31           | 400   | 400   | 372    | 334    | 289    | 252    | 210    | 170    |
| 4                                                      | 37           | 400   | 400   | 400    | 397    | 340    | 298    | 258    | 218    |
| 4.5                                                    | 43           | 400   | 400   | 400    | 400    | 400    | 397    | 331    | 281    |
| 5                                                      | 49           | 400   | 400   | 400    | 400    | 400    | 400    | 369    | 319    |
| 5.5                                                    | 55           | 400   | 400   | 400    | 400    | 400    | 400    | 400    | 400    |
| 6                                                      | 61           | 400   | 400   | 400    | 400    | 400    | 400    | 400    | 400    |

| Slab Thickness (inches) | 8'-0" | 9'-0" | 10'-0" | 11'-0" | 12'-0" | 13'-0" | 14'-0" | 15'-0" |
|-------------------------|-------|-------|--------|--------|--------|--------|--------|--------|
| 3.5                     | 177   | 156   | 135    | 121    | 101    | 85     | 70     | 57     |
| 4                       | 224   | 197   | 175    | 156    | 139    | 125    | 113    | 102    |
| 4.5                     | 273   | 241   | 215    | 190    | 170    | 153    | 138    | 126    |
| 5                       | 323   | 286   | 253    | 226    | 202    | 182    | 164    | 149    |
| 5.5                     | 375   | 331   | 294    | 262    | 235    | 211    | 191    | 173    |
| 6                       | 430   | 378   | 336    | 300    | 269    | 242    | 218    | 197    |

| 20/18/16 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi) |              |       |       |        |        |        |        |        |        |
|--------------------------------------------------------------|--------------|-------|-------|--------|--------|--------|--------|--------|--------|
| Slab Thickness (inches)                                      | Weight (psf) | 8'-0" | 9'-0" | 10'-0" | 11'-0" | 12'-0" | 13'-0" | 14'-0" | 15'-0" |
| 3.5                                                          | 31           | 400   | 400   | 400    | 390    | 337    | 285    | 246    | 207    |
| 4                                                            | 37           | 400   | 400   | 400    | 400    | 400    | 398    | 332    | 282    |
| 4.5                                                          | 43           | 400   | 400   | 400    | 400    | 400    | 400    | 381    | 331    |
| 5                                                            | 49           | 400   | 400   | 400    | 400    | 400    | 400    | 400    | 400    |
| 5.5                                                          | 55           | 400   | 400   | 400    | 400    | 400    | 400    | 400    | 400    |
| 6                                                            | 61           | 400   | 400   | 400    | 400    | 400    | 400    | 400    | 400    |

| Slab Thickness (inches) | 8'-0" | 9'-0" | 10'-0" | 11'-0" | 12'-0" | 13'-0" | 14'-0" | 15'-0" |
|-------------------------|-------|-------|--------|--------|--------|--------|--------|--------|
| 3.5                     | 236   | 181   | 170    | 151    | 136    | 122    | 110    | 100    |
| 4                       | 274   | 242   | 215    | 190    | 172    | 155    | 140    | 127    |
| 4.5                     | 315   | 276   | 253    | 226    | 202    | 182    | 164    | 149    |
| 5                       | 358   | 313   | 282    | 251    | 226    | 203    | 186    | 169    |
| 5.5                     | 400   | 364   | 335    | 305    | 279    | 253    | 236    | 219    |
| 6                       | 450   | 400   | 364    | 335    | 305    | 279    | 253    | 236    |

**Note**  
Due to the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 ga. However, the construction square do get longer for 18 and 16 ga deck.